

Step 1: Understand the relation and FDs

We are given a relation:

- $R(\text{student}, \text{teacher}, \text{subject})$

with the following functional dependencies (FDs):

- $(\text{student}, \text{teacher}) \rightarrow \text{subject}$
- $(\text{student}, \text{subject}) \rightarrow \text{teacher}$
- $\text{teacher} \rightarrow \text{subject}$

Interpretation:

- If you know which **student** is with which **teacher**, you know the **subject** they are studying.
 - If you know which **student** is studying which **subject**, you know the **teacher**.
 - If you know the **teacher**, you know the **subject** they teach.
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Step 2: Find the candidate keys

A candidate key is a minimal set of attributes that can uniquely identify a tuple.

From the FDs:

1. $(\text{student}, \text{teacher}) \rightarrow \text{subject}$
 - Together, student and teacher decide subject, so $(\text{student}, \text{teacher})$ can determine all attributes $(\text{student}, \text{teacher}, \text{subject})$.
 - Hence, $(\text{student}, \text{teacher})$ is a candidate key.
2. $(\text{student}, \text{subject}) \rightarrow \text{teacher}$
 - Together, student and subject decide teacher, so $(\text{student}, \text{subject})$ can also determine all attributes.
 - Hence, $(\text{student}, \text{subject})$ is another candidate key.

So the candidate keys are:

- $\{\text{student}, \text{teacher}\}$
- $\{\text{student}, \text{subject}\}$

Note:

- **student** alone is not a key; it cannot determine teacher or subject.
 - **teacher** alone is not a key; it only determines subject, not student.
 - **subject** alone is not a key; it does not determine student or teacher.
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Step 3: Check if the relation is in 3NF

Recall 3NF condition (informal):

- Relation must be in 2NF, and
- For every non-trivial FD $X \rightarrow Y$, either:
 - X is a superkey, or
 - Each attribute in Y is **prime** (part of some candidate key).

Check each FD:

1. $(\text{student}, \text{teacher}) \rightarrow \text{subject}$
 - Left side $(\text{student}, \text{teacher})$ is a candidate key, hence a superkey.
 - This FD is fine for 3NF.
2. $(\text{student}, \text{subject}) \rightarrow \text{teacher}$
 - Left side $(\text{student}, \text{subject})$ is a candidate key, hence a superkey.
 - This FD is also fine for 3NF.
3. $\text{teacher} \rightarrow \text{subject}$
 - Left side teacher is **not** a key.
 - Right side subject is part of a candidate key (because $(\text{student}, \text{subject})$ is a candidate key), so subject is a **prime** attribute.

Because the right-hand side is a prime attribute, this FD satisfies 3NF's second condition. Therefore:

- The relation R is in 3NF (there is no transitive dependency via a non-key attribute).
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Step 4: Check BCNF condition

BCNF is stricter than 3NF. Its condition:

- For every non-trivial FD $X \rightarrow Y$, X must be a **superkey**.

Again, check each FD:

1. $(\text{student}, \text{teacher}) \rightarrow \text{subject}$
 - $(\text{student}, \text{teacher})$ is a candidate key $\rightarrow$ superkey $\rightarrow$ satisfies BCNF.
2. $(\text{student}, \text{subject}) \rightarrow \text{teacher}$
 - $(\text{student}, \text{subject})$ is a candidate key $\rightarrow$ superkey $\rightarrow$ satisfies BCNF.
3. $\text{teacher} \rightarrow \text{subject}$
 - Left side = teacher, which is **not** a candidate key or superkey.
 - Right side = subject (non-empty).
 - This FD **violates BCNF**, because the determinant (teacher) is not a superkey.

So:

- R is in 3NF but **not** in BCNF.
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Step 5: Understand the anomaly

Because teacher is a determinant ($\text{teacher} \rightarrow \text{subject}$) but not a key, anomalies can occur

Example (from the notes):

- Suppose we have tuples for a student “Tahira” who is taught by “N.Gupta” in subject “C”.
 - If we delete the student Tahira’s row, we also lose the information that **N.Gupta teaches C**, because that info was only stored in that row. This is a **deletion anomaly**:
 - $\text{Teacher} \rightarrow \text{subject}$ is an important fact about teachers and subjects, but it is stored mixed with student data.
 - Deleting one student can accidentally delete knowledge about which subject a teacher teaches.
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Step 6: Decompose R into BCNF

To fix the BCNF violation, we decompose using the problematic FD:

- $\text{teacher} \rightarrow \text{subject}$

We create two relations:

1. R1(Teacher, Subject)
 - This stores which teacher teaches which subject.

- FD: teacher $\rightarrow$ subject.
- Here, teacher is a key in R1, so this relation satisfies BCNF.

2. R2(Student, Teacher)

- This stores which student is assigned to which teacher.
- Candidate key: (student, teacher) (assuming a student can have the same teacher at most once).
- There is no FD with a non-key determinant, so R2 is also in BCNF. After decomposition:
 - Teacher–subject information is separated into R1.
 - Student–teacher pairing is in R2.
 - The BCNF violation is removed, and anomalies like the “Tahira” deletion problem are avoided.

Step 7: Final summary

- Original relation R(student, teacher, subject) had FDs making it 3NF but not BCNF because teacher $\rightarrow$ subject with teacher not being a key violates BCNF.
- This caused anomalies: deleting a student could remove information about which subject a teacher teaches.
- Decomposition into:
 - R1(Teacher, Subject) and
 - R2(Student, Teacher)
 makes every determinant a key in its relation, so both relations are in BCNF and anomalies are removed.